Wind Farm Suitability Study in West Virginia

GP Gomez Parra

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Abstract & Keywords

West Virginia needs to look into wind energy to continue to be an energy powerhouse for the United States. This project looks at other projects using a multiple criteria decision-making process of use criteria to determine the best suitable sites for wind farms in the state of West Virginia. The criteria were broken down into three groups of environmental constraints, human acceptability, and economic viability. This project focused on giving all groups an equal weight using binary and linear weighted multiple criteria decision-making analysis. The end showed that 13% of the state of West Virginia is suitable for wind farm development, with these concentrations seen in the Allegheny Highlands and the Eastern Panhandle regions.

Keywords: West Virginia, wind farm, suitability, raster math, multiple criteria decision-making

Introduction

West Virginia is known for being an energy powerhouse, with producing much of the coal that propelled the United States onto the world stage (Ajanaku et al., 2022). Today, the state of West Virginia still gets 95% of its energy needs from coal, but with greenhouse emissions and climate change becoming a growing concern, this needs to change (Ajanaku et al., 2022). West Virginia needs to transition into green energy to remain being that energy power house moving forward. Wind energy, as explained by Rehman & et. al. (2020), “has become one of the most promising renewable, sustainable, and commercially accepted sources of energy” (Rehman et al., 2020). Wind turbines have only become more efficient with time, allowing more electricity to be produced from a single turbine (Office of Energy Efficiency & Renewable Energy, 2024). While most onshore wind potential, according to the National Renewable Energy data set, is in the Midwest and Rocky Mountains, a large section of West Virginia also has that same potential (King et al., 2014).

The decision-making process on where to develop wind farms stems from using ecological and spatial planning in GIS (Ajanaku et al., 2022). Multiple sources, see TABLE 1, use a variety of different criteria in determining where to place wind farms. These criteria are selected from three overarching groups: environmental constraints, human acceptability, and economic viability. The importance of each of these groups and their specific conditions varies across studies, due to different priorities given to each of the three groups. The environmental constraints group of criteria is those involved in the safety of the natural landscape and the impacts wind turbines can have on the environment (Braun et al., 2022). These criteria are required to be taken into account by law in the United States by several federal, state, and local laws (Braun et al., 2022). Distance from protected bird areas, distance from parks, and distance from water bodies are examples of these criteria (Braun et al., 2022). Human acceptability is the criteria that contribute to wind farm projects not being approved in the planning process (Tsani

et al., 2024). Distance from residential areas, distance from airports, and visibility of wind turbines are examples of human acceptability criteria (Ajanaku et al., 2022; Gorsevski et al., 2013; Braun et al., 2022). Economic viability are criteria that tries to limit the cost of wind farms. These criteria focus on trying to make wind farms a viable alternative to traditional energy generation, which starts with making wind farms economically competitive, in other words, as cheaply as possible (Tsani et al., 2024). Wind speed potential, slopes, elevation, distance from transmission lines, proximity to roads, and proximity to railroads are examples of economic viability criteria (Ajanaku et al., 2022; Braun et al., 2022).

Addressing problems early on to avoid headaches down the line is the overarching concept that is considered in selecting criteria to use in the suitability of wind farm placement (Braun et al., 2022). Some studies focus on giving equal importance to all criteria, allowing for unsuitable areas to be excluded from further analysis if it doesn't meet the requirements of one of those criteria (Ajanaku et al., 2022; Höfer et al., 2016). While others like Ajanaku et al. (2022) and Höfer et al. (2016) use experts to create an analytic hierarchy process in determining the weights placed on different criteria. In those projects, they communicated with experts in the field of wind energy to establish weights to place different importance on different factors they took into account with their analysis (Ajanaku et al., 2022; Höfer et al., 2016). Experts consulted in Ajanaku et al. (2022) tended to show their bias towards making economically viable wind farms, giving a higher priority to wind potential, distance to transmission lines, and proximity to roads as their main priorities (Ajanaku et al., 2022). In Höfer et al. (2016), experts focused on wind potential, distance from protected areas, and distance from urban areas (Höfer et al., 2016). Using an analytic hierarchy process gives too much power to certain criteria in determining suitability, as well as showing their preferential bias to favor factors that increase economic viability for wind farms.

Materials and Methods

If West Virginia wants to continue to be a powerhouse for the United States, it needs to know where to place its wind turbines. Determining the most suitable sites for wind farm development is the purpose of this project. The criteria used to determine the best suitable sites for wind farms for this project can be seen in TABLE 2. These were selected after comparing what other studies had used to determine their suitability studies, as seen in TABLE 1. These criteria are selected from the three overarching groups: environmental constraints, human acceptability, and economic viability.

The environmental constraints criteria group is governed primarily by federal, state, and local laws placed to protect animals and their habitats (Braun et al., 2022). The main priority is placed on the association between bird and bat mortality and wind farms (Ajanaku et al., 2022). This trend of mortality was established when technology was less developed in the field of wind turbines, which, in recent studies, shows a dramatic lowering in mortality rates (Braun et al., 2022). Regardless of the technological advancements, most developers of wind farms are still recommended to develop an “Avian Protection Plan or Bat Conservation Strategies” all to help assure that these animals are being considered in the placement of these wind farms (Braun et al., 2022). Helping to alleviate the concerns of bird and bat safety, important bird areas, labeled as Critical Wildlife Areas, are used as exclusionary areas in this study. Important bird areas were also used by Ajanaku et al. (2022) and Rehman et al. (2020) in their wind farm suitability studies. Other animals and their habitats also need to be considered, which results in exclusion of federal and state parks being placed, also done by (Ajanaku et al., 2022; Rehman et al., 2020; Höfer et al., 2016; Braun et al., 2022). Most studies seen in TABLE 1 also use this exclusion of parks along with a buffer around these areas. In this study, a 2,000 m buffer is placed around all federal and state parks, as well as a 1,000 m buffer around critical wildlife areas. These exclusionary areas can be seen in MAP 1A and 1B, respectively. Lastly, water

bodies are the last habitat that was considered in most studies. Exclusion of water bodies by themselves is not enough, because of the potential wetland and hazardous runoff from wind farm sites (Rehman et al., 2020). A buffer of some kind can be seen used in other studies, see TABLE 1, but a 1,000 m buffer is used for this study, seen in MAP 1C.

Human acceptability is a factor that impacts humans from accepting wind turbines, which was cited as the main reason why new wind farm projects are rejected (Tsani et al., 2024). One of these factors is the visual impact of these wind turbines on the landscape (Braun et al., 2022). Some studies tried limiting the placement of wind turbines on summits, regardless of their increased economic potential, to try to appease the human acceptability factor (Baban & Perry, 2001). This study did not address those limitations of summits, given the varying sight lines that impact whether these turbines can be seen or not (Tsani et al., 2024). Regardless of summits, limiting the proximity of wind turbines from residential areas was placed in every study analysis, as seen in TABLE 1. Some studies use local policies to guide their buffer distances, like Rehman et al. (2020), while others base their buffer size on the zoning type of the area being buffered (Höfer et al., 2016). This study placed a buffer along with a linear weighted scale on distance from urban areas, seen on MAP 2A and 2B. These urban areas were selected from the Land Use Land Cover Map created by NAIP of 2016 (WV Division of Natural Resources, Wildlife Resources, 2020). Noise is a concern that people raise when it comes to wind farm developments (Braun et al., 2022). While there have been extensive studies on noise and infrasound, low-frequency (inaudible) sound, which are created by wind turbines, no negative impacts on the body have been seen in any of the studies (Braun et al., 2022). Regardless, there are still those who say that wind turbines, through infrasound, have a negative impact on the body (Braun et al., 2022). For the purpose of this study, noise was not considered, apart from the buffer around urban areas. Lastly, airports, while not considered by every study looked at, vary greatly in their buffers from 3, 000 m to 5,000 m (Ajanaku et al., 2022; Rehman et al.,

2020). While there is more nuance that can be taken into a wind farm placement due to elevation and flight paths for buffers around airports, this study will use the general rule of 3.5 miles (~5,600 m) recommended by the Federal Aviation Administration as its buffer for this analysis, seen in MAP 1D (Gleason, 2024).

Economic viability is the last section that needs to be considered for this study. This deals with trying to make the placement of the wind farm in the easiest location to earn the most money (Höfer et al., 2016). Wind turbines need wind to make power; thus, higher wind speeds are recommended for wind farm placement (Office of Energy Efficiency & Renewable Energy, 2024). Even with technological advancements in wind turbines, being able to produce similar amounts of energy at lower wind speeds, it is still preferable for wind farms to be in areas of at least 5 m/s wind speeds (Office of Energy Efficiency & Renewable Energy, 2024). These 5 m/s or greater wind speeds were used in 4/5 of the studies looked at, seen in TABLE 1, resulting in a weighted wind speed seen in MAP 3B used for this analysis. Another factor that impacts cost is the initial construction for site preparation for wind farms. Groundwork needed to grade for the foundations for the wind turbines is a loss of money, making wind turbines on steeper slopes less preferable (Rehman et al., 2020; Höfer et al., 2016; Baban & Perry, 2001). Slopes deemed too much, vary greatly from 30% to 10%, but given the mountainous terrain of West Virginia, a maximum slope of 15% was given for this study, seen in MAP 3A (Höfer et al., 2016; Baban & Perry, 2001). Proximity to roads, railroads, and electrical lines all creates expenses the further they are from new wind turbines. Electrical transmission lines are preferably closer to new wind farms to be able to lessen the expense of new transmission lines to connect to the greater system (Rehman et al., 2020; Höfer et al., 2016). Regardless of needing transmission lines to be close, there is such a thing as too close, with your average wind turbine having a diameter span of 133.8 m, a buffer around transmission lines was created, along with a weighted proximity seen in MAP 4C (Office of Energy Efficiency & Renewable Energy, 2024). The

diameter of the wind turbine buffer was also used when it came to proximity to railroads. While few studies took into account railroads, preferring roads, a weighted proximity was used for this study, as seen in MAP 4A. Railroads, like roads, can be used to transport wind turbine parts, which are quite large and cumbersome to transport (Office of Energy Efficiency & Renewable Energy, 2024). Lastly, all roads are not created equal in their viability for transporting wind turbine parts (Höfer et al., 2016; Gorsevski et al., 2013; Office of Energy Efficiency & Renewable Energy, 2024). While there is new technology that can allow for wind turbine parts to be created onsite, these practices are not in wide use, and thus, road proximity is used in this study, seen in MAP 4B (Office of Energy Efficiency & Renewable Energy, 2024). The proximity analysis has a cut-off at 10,000 m, seen in almost all studies, see TABLE 1, given the added cost for building roads for maintaining and transporting materials (Office of Energy Efficiency & Renewable Energy, 2024).

Now that all criteria were selected, a way to combine them together needed to be considered. A multiple criteria decision-making analysis has been used in all the studies seen in TABLE 1. Multiple criteria decision-making is used to make all factors have equal importance when siting wind farm suitability (Ajanaku et al., 2022; Höfer et al., 2016). Some factors, as seen in MAP 1, only have a weighted preference of being in a location or not in a location. The remaining factors have a linear weighted preference regarding their location, seen in TABLE 2 and Map 2-4 (Gorsevski et al., 2013; Tsani et al., 2024). This analysis gives all factors equal weight when determining wind farm placement (Baban & Perry, 2001). An analytic hierarchy process places a weight on different criteria, giving those factors greater pull over others (Ajanaku et al., 2022; Höfer et al., 2016). Weighting factors is difficult for experts to agree on and typically prioritize economic viability over all other factors, which is why this analysis was not used in this study (Tsani et al., 2024). The final combination of the criteria was done by transforming the binary criteria factors, seen in MAP 1, scaled to 0 and 100. The remaining

factors were weighted linearly by proximity, with their weights seen in MAP 2-4. These factors were then combined using the raster calculator, with the highest values being preferable to lower values. The final result of suitability can be seen in MAP 5.

Results and Discussion

After combining all the factors seen in TABLE 2 and MAP 1-4, the best suitability for wind farms in West Virginia can be seen in MAP 5. MAP 5 shows the two concentrations of excellent suitability in West Virginia seen in Allegheny Highlands and the Eastern Panhandle, respectively (West Virginia Geological and Economic Survey, 2020). In the state as a whole, in this this analysis shows that 514 mi^2 were deemed to have excellent suitability for wind farm potential, with the number for very good and good jumping to 1,118 mi^2 and 974 $mi^{2,}$ respectively. This means that roughly 13% of the state of West Virginia is suitable for wind farm development.

Only one other paper has done a similar wind farm suitability on West Virginia (Ajanaku et al., 2022). Ajanaku et al. (2022)'s criteria can be seen in TABLE 1. They used an analytic hierarchy process based on experts that prioritized wind potential, distance to transmission lines, and proximity to roads (Ajanaku et al., 2022). This study just used a multiple-criteria decision-making analysis with binary and linear proximity weights for all criteria being used. Regardless of these differences, it is surprising to see that there were still counties that were deemed suitable in both studies. In Ajanaku & et al (2022), the counties of Tucker, Grant, Mineral, and Pocahontas were shown as being the best for wind farm development (Ajanaku et al., 2022). In comparison, this study shows that the counties of Grant, Mineral, Upshur, Barbour, Preston, and western Randolph counties show the best wind farm development potential. Meaning that both Grant and Mineral counties were deemed suitable by both studies. Going further, placing existing wind farms on this analysis shows some wind turbines are in suitable areas, while some are in unsuitable areas, as seen in Map 6.

Conclusions

Placing existing wind turbines on this suitability map, seen on MAP 6, showcases some limitations when it comes to the analysis done. It is clear to see that not all criteria were deemed as important when creating these established wind farms. This could be due to different priorities being placed, as seen in Ajanaku et al. (2022) and Höfer et al. (2016), with their experts prioritizing economic viability over other factors. Regardless, this only helps highlight that using these broad overarching criteria might help in determining where wind farms can be placed generally; there still needs to be a site-by-site study for wind farm development (Höfer et al., 2016).

One of the biggest challenges in creating wind farm suitability studies is the integration of social factors into the multiple criteria decision-making process (Tsani et al., 2024). As stated several times in this paper, human opposition is often cited as the main reason for a wind farm project from being denied (Tsani et al., 2024). There needs to be a better way to get local input in wind farm projects, along with educating the public to the benefits of wind farm developments (Tsani et al., 2024). One factor that helps the public approve wind farms is placing wind farms on land use that was deemed undesirable, like landfills or strip mines (Höfer et al., 2016).

The next steps for the state of West Virginia, given this study and Ajanaku et al. (2022), would be to look into wind farm developments in the counties of Grant and Mineral, which both studies said were suitable. A further study on a county level could be done, including land use and zoning information, to help guide the state of West Virginia to continue to be the powerhouse of the United States.

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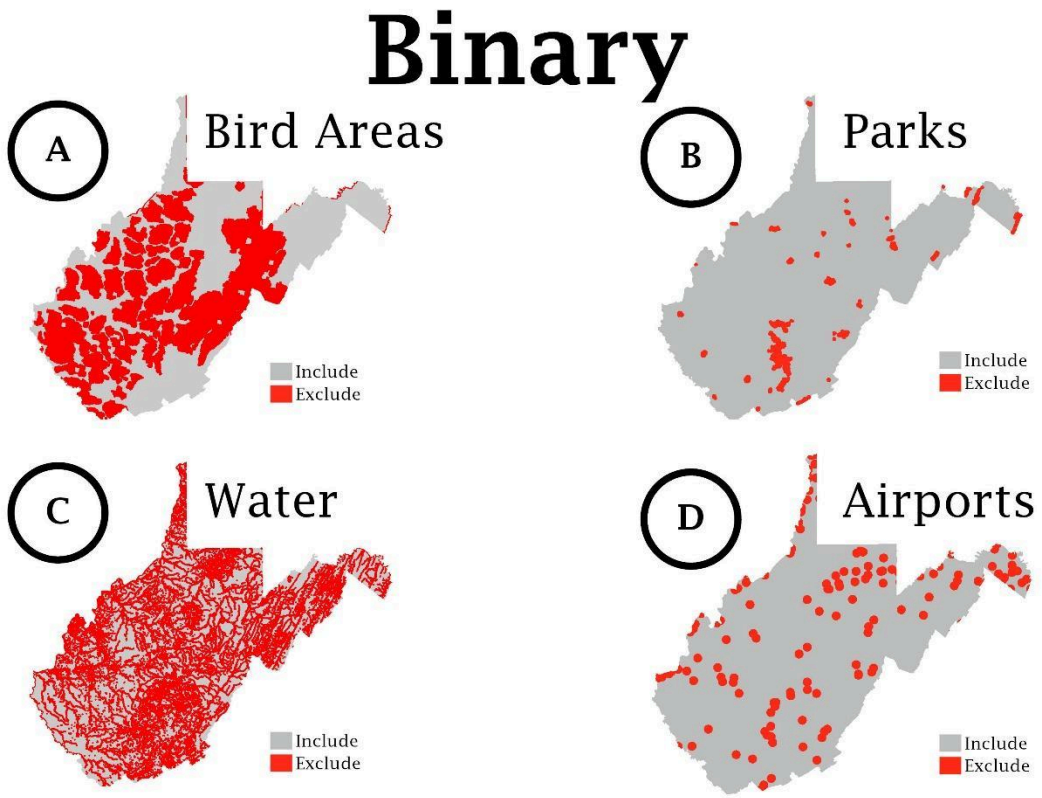
Tables and Maps

Table 1. Comparing Factors for Wind Farm Siting. Each wind farm suitability study has different criteria used for its suitability study; this shows a summary of these criteria. Blank areas are those where these factors were not considered in that study.

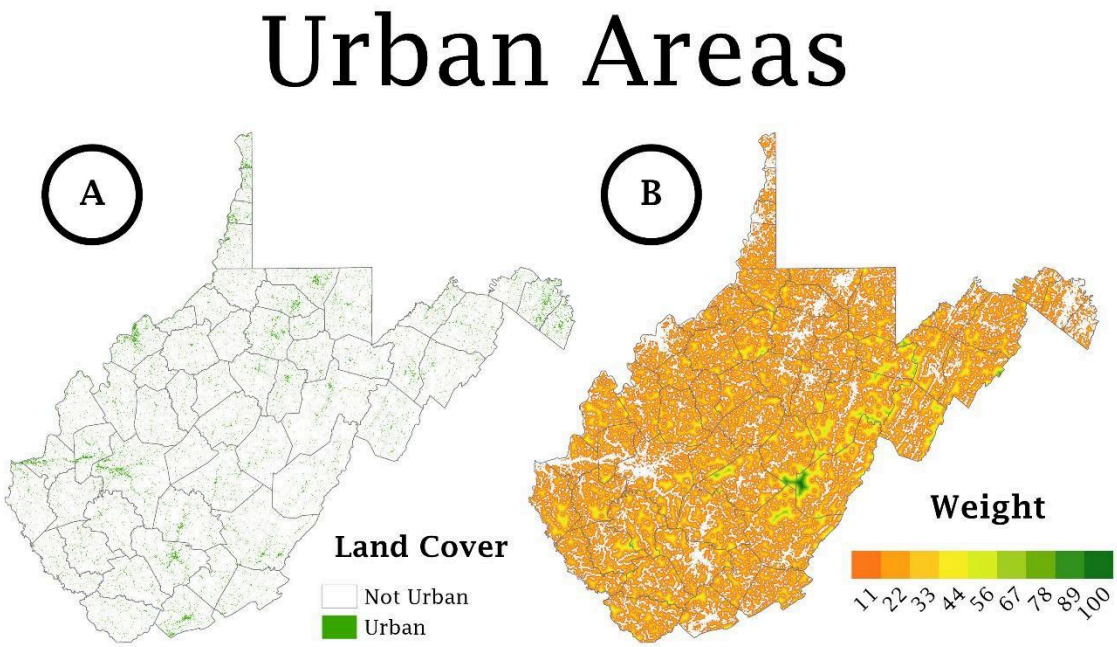
Criteria	Ajanaku et al. (2018)	Rehman et al. (2020)	Höfer et al. (2016)	Baban & Perry (2001)	Gorsevski et al. (2013)
Wind Potential	> 5 m/s	> 5.8 m/s	> 6 m/s	> 5 m/s	Fuzzy
Slope	< 10%	< 15%	< 30%	< 10%	
Elevation	< 2,000 m				
Direction of Slope	Western			Western	
Transmission Lines	< 10,000 m	< 10,000 m	> 100 m	< 10,000 m	1,000 m – 20,000 m
Roads	< 10,000 m	1,000 m – 10,000 m		< 10,000 m	1,000 m – 10,000 m
Railroads			> 100 m		1,000 m – 10,000 m
Urban Areas	> 500 m	2,000 m – 15,000 m	> 550 m	> 2,000 m	Density Fuzzy
Airports	> 3,000 m	> 5,000 m			
Parks	> 2,000 m	> 2,000 m	> 300 m	> 500 m	
Critical Wildlife Areas	> 500 m	> 1,500 m			
Water Bodies	> 2,000 m	> 1,000 m	> 50 m	> 400 m	

Table 2. Factors and Evaluation Methods Used for This Analysis. Based on other studies, this table shows the criteria used, their minimum and maximum values, the weighted method, values that are perceived as more desirable, intervals used for the weights, and the number of weights created for that criterion.

Criteria	Minimum	Maximum	Method	Desire	Intervals	Weights
Wind Potential	5 m/s		Linear	Higher	1 m/s	7
Slope		15%	Linear	Lower	2.5%	6
Transmission Lines	300 m	10,000 m	Linear	Lower	1,000 m	10
Roads		10,000 m	Linear	Lower	1,000 m	9
Railroads	1,000 m	10,000 m	Linear	Lower	1,000 m	9
Urban Areas	500 m		Linear	Higher	1,000 m	9
Airports	5,000 m		Binary	No		
Parks	2,000 m		Binary	No		
Critical Wildlife Areas	1,000 m		Binary	No		
Water Bodies	1,000 m		Binary	No		

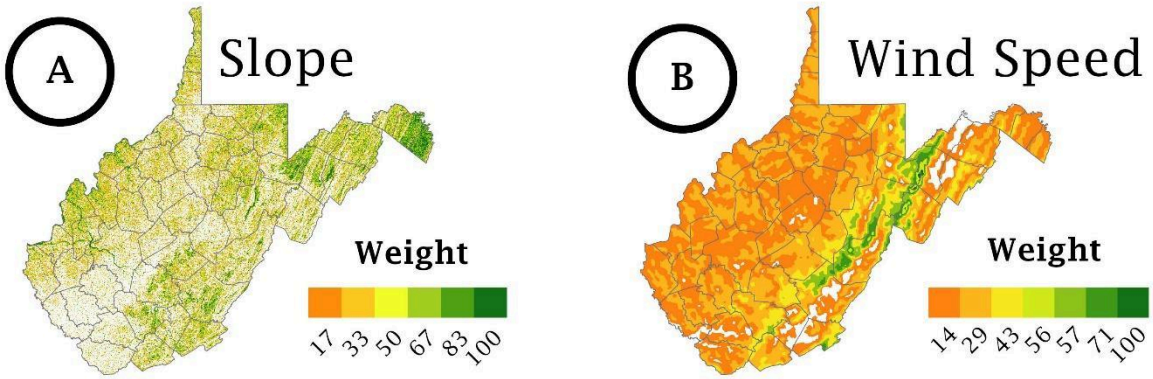


Map 1. Binary Criteria. (A) Bird areas, **(B)** parks, **(C)** water bodies, and **(D)** airports were all weighted binarily. These maps show areas that were binary in their weights, with areas to be included weighted 100 and areas excluded weighted 0.

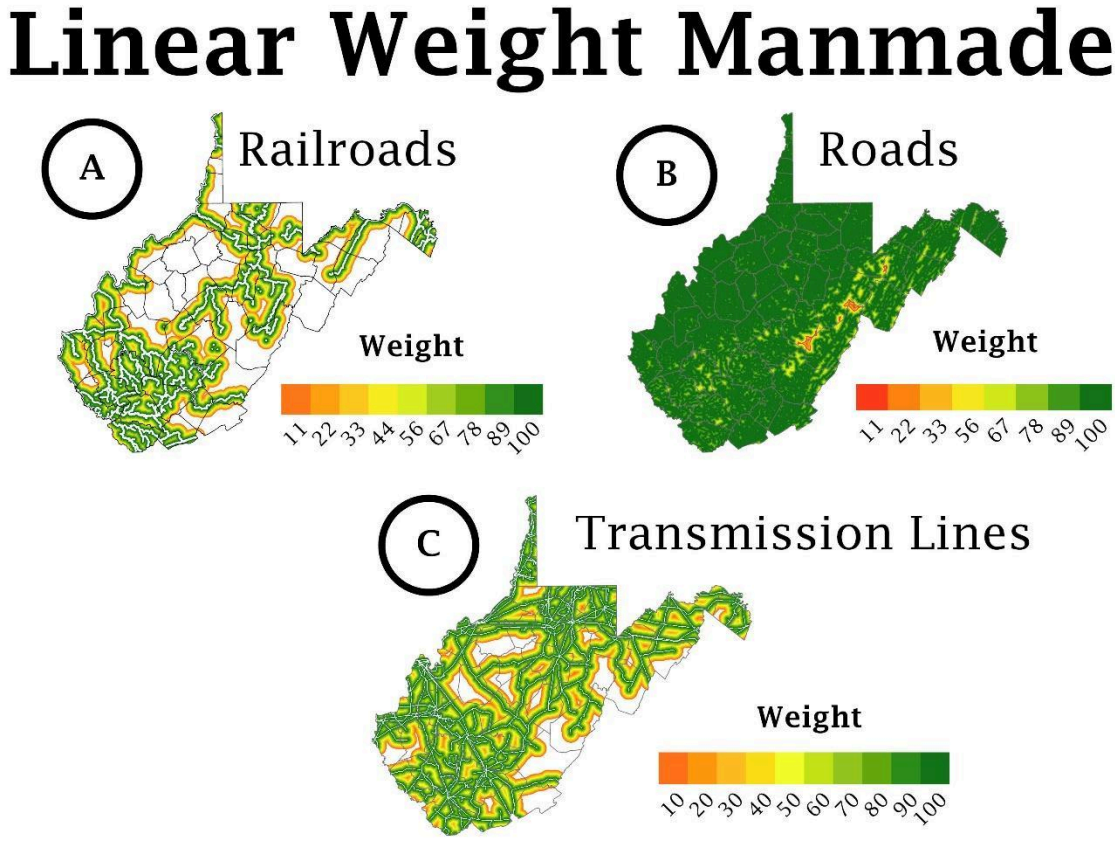


Map 2. Urban Areas. (A) Urban areas were selected from NAIP LULC 2016 Map. **(B)** Then, a buffer followed by a linear weighted buffer around the urban areas was performed.

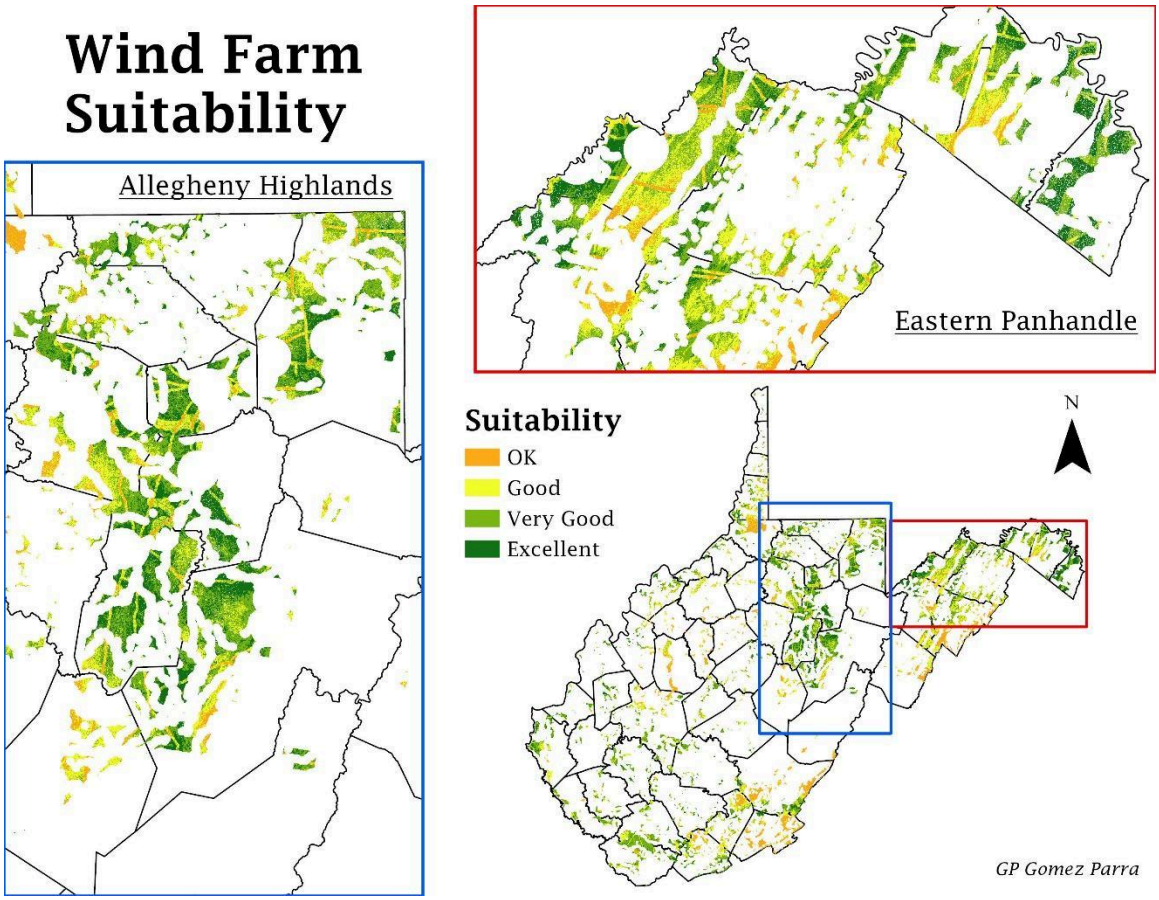
Linear Weight Natural



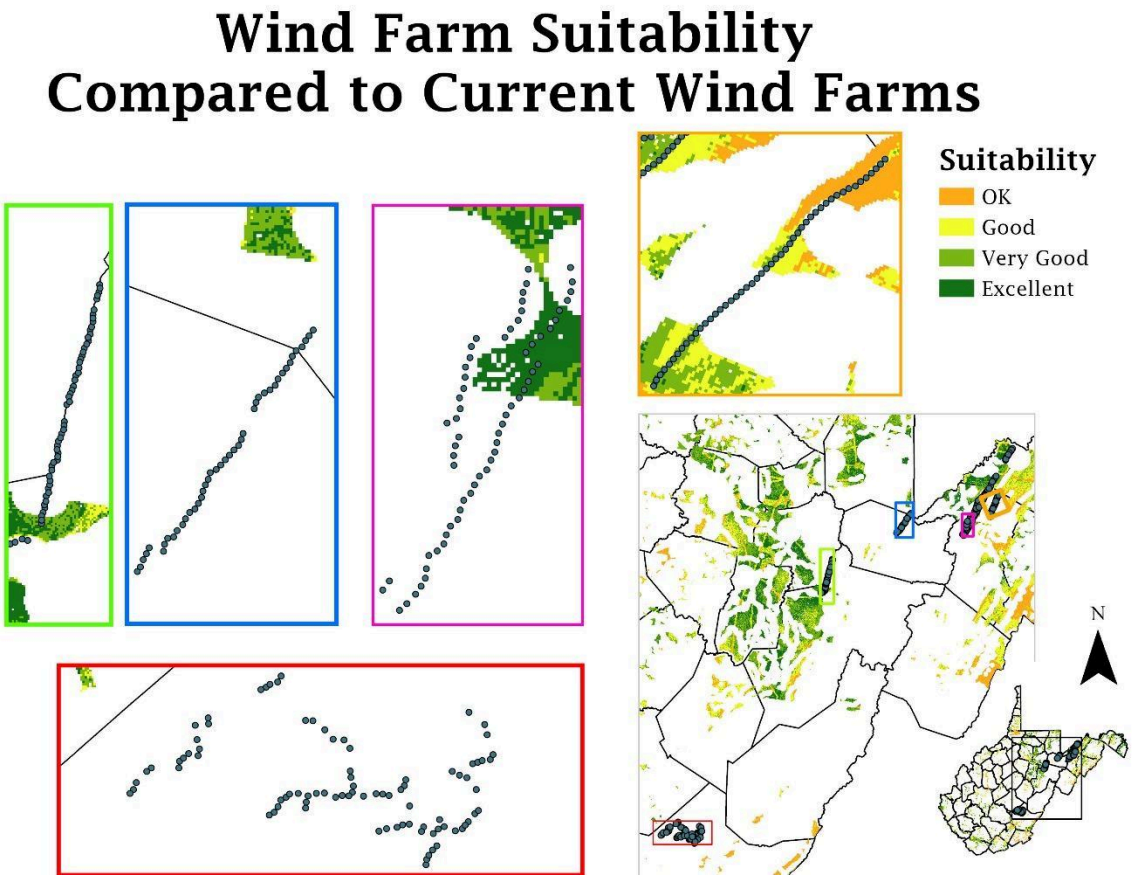
Map 3. Linear Weighted Natural. (A) Slope of >15% excluded, the remaining slope values were linearly weighted, with increasing values closer to 0%. **(B)** Wind speeds of <5m/s were excluded, with the remaining values linearly weighted, with increasing values the higher the wind speeds.



Map 4. Linear Weighted Manmade. (A) Railroad proximity linearly weighted, with increasing values closer to the buffer around the railroad tracks of 1,000 m to account for the diameter of wind turbine blades and wind created by trains. **(B)** Road proximity is linearly weighted, showing increasing values closer to the roads they got. **(C)** Transmission lines are costly, with increasing values closer to the buffer round the lines around the transmission lines or 300 m to account for the diameter of wind turbine blades.



Map 5. Wind Farm Suitability. Results of the final analysis of the multiple criteria decision-making process. **(Blue)** Shows the concentration of excellent suitability seen in the area termed the Allegheny Highlands. **(Red)** Shows the concentration of excellent suitability seen in the area known as the Eastern Panhandle of West Virginia.



Map 6. Wind Farm Suitability Compared to Current Wind Farms. Results of the final analysis of the multiple criteria decision-making process overlaid with the current wind farms in the state of West Virginia. There are four main constraints of wind farms in the state, with all but one area falling mostly within non-suitable locations.